Class Test 2

Electricity and Electronics EBN 122

7 October 2022

Test Information			
Maximum marks:	16	Full marks:	15
Duration of test:	35 min	Open/Closed book:	Closed
Total number of pages (this page included)		7	

Lecturers: Prof JP de Villiers, Prof JP Jacobs

INSTRUCTIONS

Do Step 0 in the first 5 minutes of the session. (-5 to 0 min)

Do Step 1 in the first 30 minutes of the test. (0-30 min)

Do Step 2 within the following 5 minutes. (30-35 min)

STEP 0 (-5 min to 0 min): Complete the front page of the OCR answer sheet.

STEP 1 (0 min to 30 min): Answer all questions in the question paper. Do your calculations in the spaces provided on the question paper.

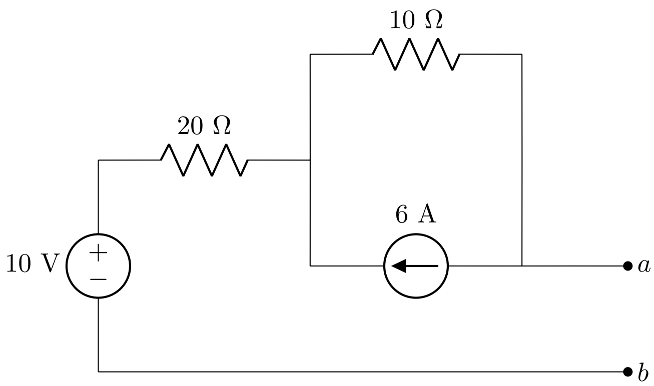
STEP 2 (30 min to 35 min): Transfer your answers to the OCR answer sheet, including units where applicable.

- The Assessment ID is: **2022EBN122C02**
- Read the detailed instructions on the front page of the OCR answer sheet as to how characters should be entered on the sheet.
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

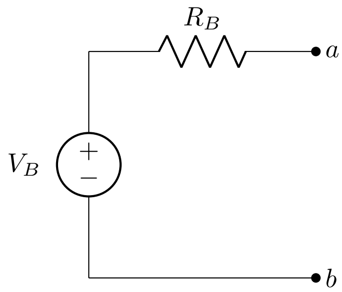
Section 1

Question 1 [2]
Use source transformation to obtain Circuit B so that it is equivalent to Circuit A with respect to terminals $a - b$.

Circuit A:



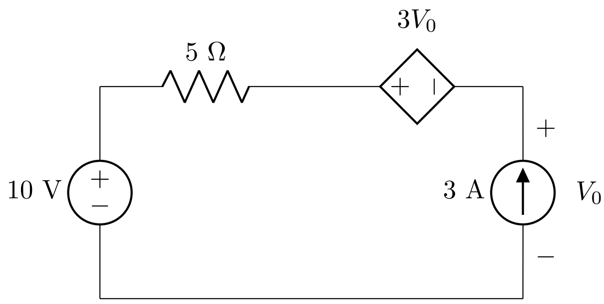
Circuit B:



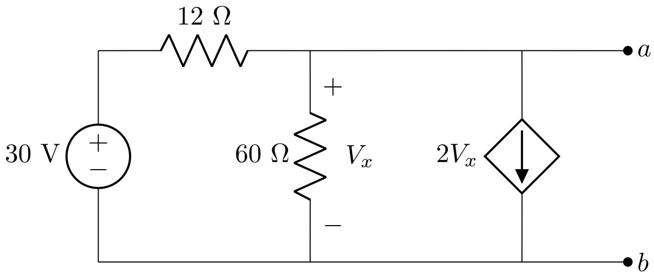
01	Calculate the value of the voltage source V_B. [2]

Questions 2 to 3 [2]

Calculate the independent contributions of the 10 V and 3 A sources to the voltage V_0.



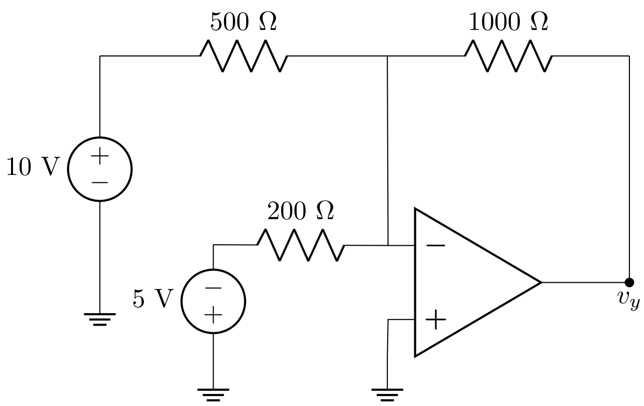
02	Contribution to V_0 of 10 V source. [1]
03	Contribution to V_0 of 3 A source. [1]

Questions 4 to 5 [4]	
For the circuit shown, find the Thevenin equivalent voltage V_{Th} and Thevenin equivalent resistance R_{Th} with reference to terminals $a - b$.	
	
04	V_{Th} [2]
05	R_{Th} [2]

Total Section 1: [8]

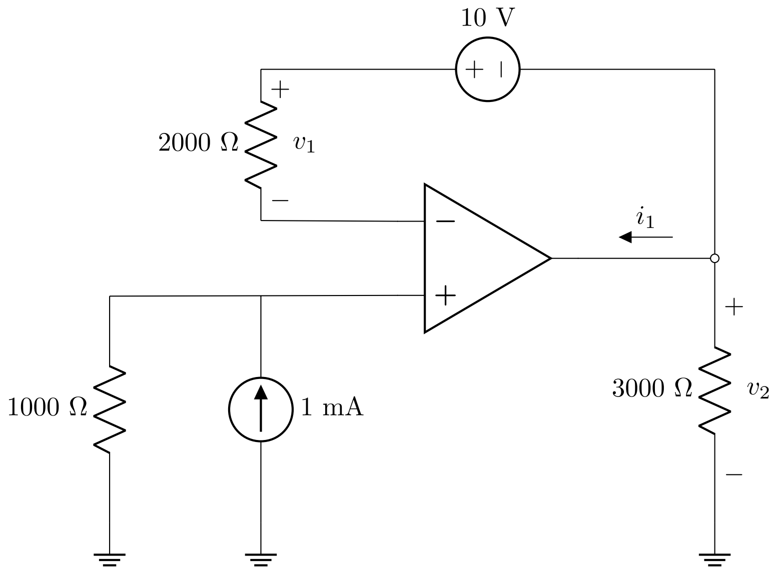
Section 2

Question 6 [2]



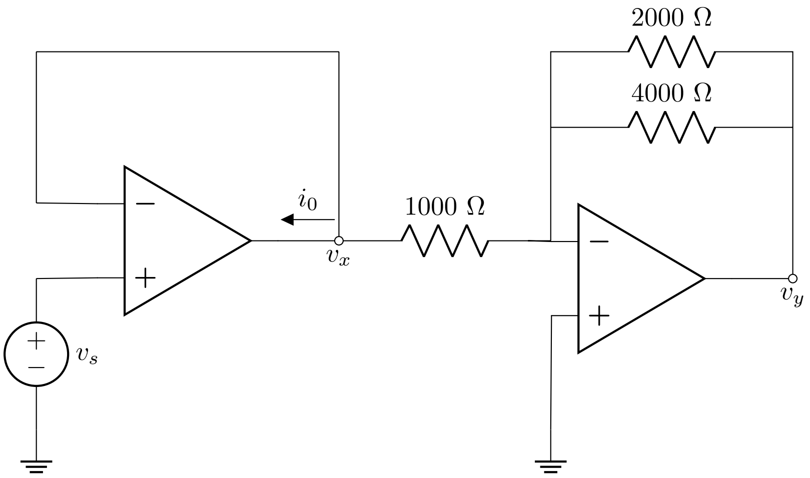
06	Determine v_y. [2]

Questions 7 to 9 [3]



07	Determine v_1. [1]
08	Determine v_2. [1]
09	Determine i_1. [1]

Questions 10 to 12 [3]	
Given: $i_0 = 2$ mA.	



10	Determine v_x. [1]
11	Determine v_y. [1]
12	Determine v_s. [1]

Total Section 2: [8]

Grand Total : [16]